

Mahagama Ralalage Dilusha Heshan Hemachandra

Mechanical Engineering Undergraduate

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Profile / Summary

Mechanical Engineering undergraduate with hands-on experience in mechanical design, CAD modelling, prototyping, mechanism design, product development, and automation. Skilled in SolidWorks, AutoCAD, mechanical assemblies, design validation, and basic FEA/CFD, with knowledge of embedded systems, electronics, robotics, and automated systems. Seeking a Mechanical Engineering internship in product development, R&D, automation, and manufacturing-oriented projects.

Education

University of Moratuwa, Sri Lanka **2024 – Present**
B.Sc. Engineering in Mechanical Engineering
CGPA: 3.09/4.00 Semester 5

Pinnawala Central College, Sri Lanka **2020 – 2023**
G.C.E. Advanced Level, Physical Science Stream – Combined Mathematics, Physics (A), Chemistry (A)
Z-Score: 2.2548 | Island Rank: 532

G.C.E. Ordinary Level Examination

8 A passes and 1 B pass

Technical Skills

Mechanical Design	SolidWorks, AutoCAD
Simulation & Analysis	MATLAB, Webots, ANSYS (Basic FEA & CFD)
Programming & ML	Python, C++, Machine Learning
PCB Design	Altium Designer
Embedded & Robotics	ESP32, LoRa, Raspberry Pi, Arduino
Industrial Automation	Basic PLC Programming
Manufacturing	Rapid Prototyping, Design for Manufacture
Languages	Sinhala (Native), English (Working Proficiency)

Selected Projects

Pattiya – Smart IoT Cattle Collar **2026 – Present**
Team Innovation Project | AgriTech, IoT and Product Development *Semi-Finalist – SPARK Challenge 2026*

- Designed and refined a weather-resistant CAD enclosure and internal component layout for a wearable cattle-monitoring device, considering sensor positioning, electronics integration, protection, and manufacturability.
- Supported integration of ESP32-based sensors and LoRa/LoRaWAN communication between the collar, environmental station, and farmer-facing monitoring system.
- Contributed to machine-learning development for analysing multi-sensor data related to estrus, heat stress, and cattle-health monitoring.

Tech Stack: SolidWorks, Altium Designer, ESP32, LoRa, Raspberry Pi, Python, Machine Learning

RoboRoarZ PathFinder – Autonomous Maze-Solving Robot **2026**
Team Competition Project *Champion – RoboRoarZ Sri Lanka 2026*

- Co-developed an autonomous differential-drive maze-solving robot and contributed to chassis integration, sensor placement, motor mounting, and electronics integration.
- Implemented and tested embedded control functions for wall detection, turning, and cell-to-cell navigation.
- Performed iterative testing and tuning to improve motion accuracy and reliability, contributing to the team winning the RoboRoarZ Sri Lanka 2026 championship.

Tech Stack: C++, Arduino, SolidWorks

Selected Projects (Continued)

Reverse Engineering of a Motorcycle Helmet – EPS Liner Analysis

2026

Academic Design Project

Mechanical Design and Product Analysis

- Disassembled and dimensionally inspected a motorcycle helmet to understand the relationship between the outer shell, comfort padding and expanded-polystyrene impact liner.
- Reconstructed key helmet and EPS-liner geometry as detailed CAD models using measured dimensions and sectional features.
- Examined liner thickness, segmentation and energy-management features, documenting observations related to impact protection and manufacturability.

Tech Stack: SolidWorks, AutoCAD, Reverse Engineering, Dimensional Metrology

Team Bumblebee – SLRC 2026 Autonomous Dual-System Robot

2026

Team Competition Project | Simulation and System Integration

Finalist – Sri Lankan Robotics Challenge 2026

- Collaborated on a dual physical/virtual autonomous system designed to operate under real-time competition constraints and changing obstacle conditions.
- Supported robot simulation and integration of the physical platform with the Ares virtual robot.
- Contributed to an API-based communication workflow that transferred decoded coordinates and commands for parallel navigation and dynamic obstacle handling.

Tech Stack: Python, Raspberry Pi, Arduino, Webots

Micromouse PathFinder Robot

2025

Team Competition Project

Finalist – SLIIT ROBOFEST 2025

- Designed and built an autonomous micromouse platform for maze exploration, wall detection and shortest-path solving.
- Integrated distance sensors, motor drivers and embedded control electronics within a compact mobile-robot chassis.
- Implemented flood-fill path-planning logic and tuned motion for reliable straight-line tracking, repeatable turns and route optimisation.

Tech Stack: C++, Arduino, SolidWorks, Altium Designer

Smart Wall Painting Robot – Developed an automated robotic system for wall-painting applications, combining mechanical design, motion control and hardware–software integration to improve painting consistency and reduce manual effort. The project provided hands-on experience in robot fabrication, function development, programming and embedded hardware integration.

Tech Stack: SolidWorks, Arduino, C++

Ongoing Projects

Fish Feed Extruder Gearbox Design – designing and developing a multi-speed gearbox with forward and reverse gear arrangements to provide suitable speed and torque for fish-feed extrusion, while enabling controlled screw reversal for blockage recovery and maintenance.

UAV Self-Leveling Landing Gear – designed and optimized a multi-link passive landing gear mechanism to maintain stable, near-vertical foot motion and adapt to uneven terrain during landing.

SmartSole – Diabetic Foot Pressure Monitoring and Offloading Insole System (*Team Bionix*) – a 3D-printed TPU insole with embedded FSR pressure sensors and an ESP32-based Bluetooth app that passively redistributes plantar pressure and actively monitors high-risk zones to help prevent diabetic foot ulcers.

Achievements

- **Champion** – RoboRoarZ Sri Lanka International Robotics Competition, Team PathFinders, 2026
- **Finalist** – Sri Lankan Robotics Challenge (SLRC), Team Bumblebee, 2026
- **Semi-Finalist** – SPARK Challenge 2026, Team Pattiya, University of Moratuwa
- **Finalist** – SLIIT ROBOFEST 2025, University Category – Micromouse

Experience

Mechanical Design Engineer – Robonics

2026 – Present

Part-time

- Designing mechanical components and assemblies for robotics and automation products, from concept through prototyping.
- Collaborating with cross-functional teams on product development and design validation.

Reference

Dr. Dumith Jayathilaka

Senior Lecturer,

Department of Mechanical Engineering,

University of Moratuwa, Sri Lanka.

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